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Open monitoring meditation alters the EEG gamma coherence in experts meditators: The expert practice exhibit greater right intra-hemispheric functional coupling

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A R T I C L E I N F O

Keywords:
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EEG gamma coherence
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A B S T R A C T

Aim: This study investigated the differences in frontoparietal EEG gamma coherence between expert meditators (EM) and naïve meditators (NM).
Material and methods: This is a cross-sectional study with a sample of twenty-one healthy adults divided under two groups (experts meditators *vs.* naive-meditators), with analyzing the intra-hemispheric coherence of frontoparietal gamma oscillations by electroencephalography during the study steps: EEG resting-state 1, during the open presence meditation practice, and EEG resting-state 2.
Results: The findings demonstrated greater frontoparietal EEG coherence in gamma for experts meditators in the Fp1-P3, F4-P4, F8-P4 electrode pairs during rest 1 and rest 2 ($p \leq 0.0083$). In addition, we evidenced differences in the frontoparietal EEG coherence for expert meditators in F4-P4, F8-P4 during the meditation ($p \leq 0.0083$).
Conclusion: Our results can support evidence that the connectivity of the right frontoparietal network acts as a biomarker of the enhanced Open monitoring meditation training.

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1. Introduction

Open monitoring meditation styles of practice exist in different traditions of Tibetan Buddhism (Mahāmudrā and Chagchen (Tibetan, phyag chen) and Dzogchen (Tibetan, Rdzogs chen)) and currently in the Mindfulness (Ferrarelli et al., 2013; Tanaka et al., 2015). The attentional focus permits the individual perceptual experiences of the consciousness (auditory and other sensory input, as well as your thoughts and emotions) to become part of your awareness and then fade away (Tanaka et al., 2015). Thus, the open-monitoring meditation is a set of regulatory and self-inquiry mental training regimes cultivated for various ends, including well-being and psychological health (Ferrarelli et al., 2013; Kummar, Correia, and Fujiyama, 2019). Studies into the open presence meditation area have had a steep growth trajectory in recent decades. Due to this, there is a growing body of evidence for the efficacy of various mindfulness-based interventions in a broad range of populations, be it to improve general well-being amongst healthy individuals or in the treatment of neurological disorders (Chiesa and Serretti, 2009; Hofmann et al., 2010; Eberth and Sedlmeier, 2012; Khoury et al., 2013).

The continuous practice of the open monitoring meditation demonstrates an evolution of cognitive skill and neural plasticity (Tang, Hölzel, and Posner, 2015) since it alters the connectivity of frontoparietal neural networks during the experience of attentional focus (Chambers, Gullone, and Allen, 2009; Hölzel et al., 2011; Hasenkamp et al., 2012; Malinowski, 2013). Like any skill, the development of mindfulness starts as a deliberate, effortful process to focus attention and, with practice, becomes less effortful as skills are embedded (Brefczynski-Lewis et al., 2007; Babiloni et al., 2010; Tang, Hölzel, and Posner, 2015; Russell, 2017). The investigation of cortical activity with electroencephalography (EEG), such as EEG resting state or during practice, is a sensitive approach to identifying these changes in neurophysiological activity. Some studies found meditation-related increases in alpha and theta power, particularly in the frontal cortex, associated with an increased sense of relaxation (Cahn, Delorme, and Polich, 2010). Electrophysiological studies have focused on the low frequency theta and alpha power changes, suggesting lower frontal activity during the practice (Kerr et al., 2013; Lomas, Ivtzan, and Fu, 2015).

These neuronal changes are related to modulations during affective and cognitive behavior in task execution (Ferrarelli et al., 2013). While these networks are engaged during daily activities, there are types of meditation to train the ability to be more aware of this whole cycle of brain activity. Focused attention practices require an intentional and sustained focus on a single object (e.g. the breath) (Bishop et al., 2004; Jo et al., 2015). Alternatively, open monitoring practices have a wider aperture of attentional focus and require "light-touch" attention to everything that arises, whether an internal stimulus such as a thought or something external, such as a sound (Lutz et al., 2008). These different practices engage the same networks but at different frequencies depending on the task.

Few studies investigated the EEG gamma coherence to investigate the communication between areas (Cahn, Delorme, and Polich, 2013; Dentico et al., 2016). In this context, past studies suggest that expert meditators had an EEG gamma power increase in frontal and parietal-occipital in the expert meditators during open monitoring meditation (Lutz et al., 2004; Cahn, Delorme, and Polich, 2010). In a previous EEG study, meditation experts showed higher parietal-occipital gamma power than novices during resting state preceding meditation practice (Berkovich-Ohana, Glicksohn, and Goldstein, 2012).

An important question raised is whether EEG gamma coherence can be related to the frontoparietal network for spatial attention processing and conscious perception associated with high-level cognitive function (Chica and Bartolomeo, 2012). In this study, we hypothesized that more meditation practice years are associated with more right intra-hemispheric functional coherence (this may imply a shift toward more effective task preparation and task switching in meditators, which one would expect to be beneficial in many executive tasks). Consistent with this interpretation are reports of meditators performing better in different attention and executive function tasks, which are directly related to greater activation of the right hemisphere (Florian Kurth et al., 2015; Davis and Hayes, 2011).). In addition, the reported effortless right intra-hemispheric functional coupling between brain states in experienced meditators may be partly due to an effective task preparation. Previous studies demonstrated that long-term meditation practice increases the connectivity between the areas of the frontoparietal network, leading to working memory efficiency (Hauswald et al., 2015; Dor-Ziderman et al., 2013; Langer et al., 2013), sensorimotor integration (Kumari et al., 2017), and the ability to inhibit distractors (Babiloni et al., 2006; Bendat and Allan, 2011).

The central aim of EEG analysis is to relate measures of neural dynamics to behavior or cognition. One of the most promising measures of such dynamics is coherence, a correlation coefficient (squared) that estimates the consistency of relative amplitude and phase between any pair of signals in each frequency band (Bendat and Allan, 2011). EEG coherence provides a critical estimate of functional interactions between cortical areas. EEG coherence may yield information about network formation and functional integration across brain regions. EEG coherence may underlie processes that bind together internal sensations and external stimuli and, in the case of open monitoring, may support the ability to notice and integrate distractors while effortlessly maintaining relaxed attention to whatever arises in the space of the mind (Dentico et al., 2016; Scheibner et al., 2017; Hudak et al., 2021). The aim of the current study is to investigate the differences in the Intra-hemispheric coherence of frontoparietal gamma oscillations by electroencephalography between experts and naive meditators.

2. Material and methods

2.1. Participants

We recruited twenty-one healthy adults (12 men and 9 women), out of which eleven were expert meditators (EM) (mean age 43.8 $\pm$ 17.53), and ten were naive-meditators (NM) (mean age 40.10 $\pm$ 14.72). Only right-handed individuals were selected based on the Edinburgh Inventory (Oldfield, 1971). All participants reported moderate consumption of alcoholic drinks and coffee. Participants did not use any substance that could influence brain activities (e.g., tobacco, coffee, alcoholic beverages, caffeine-containing foods, or medications) 24 h before or during the study period. All participants underwent a medical evaluation to exclude neurological or motor diseases, visual impairment, hearing, cognitive, and motor impairment that would impair the performance of the meditation.

1) The EM group included monks (long-term meditation practitioners) with mean meditation of 12.23 ± 7.65 years, and living in the same city, under the same socio-economic conditions and exposed to the same environmental stimuli as the control group (i.e., they were not living in isolated in Buddhist centers). The group was selected based on a criterion of at least ten years of formal meditation practice in the Nyingma, Zen, Tibetan, Vipassana, Kagyu traditions of Tibetan Buddhism, which have very similar practice styles. A person must have at least five years of meditation practice to be considered an expert. The length of their training was estimated based on their daily practice and the time (ten hours of sitting meditation per day) they spent in meditative retreats. Based on this criterion, these practitioners are referred to as "experts" here (Fucci et al., 2018). Regarding the sample homogeneity to the formal meditation practice, sample *t*-test in the EM group demonstrated no significant difference in meditation time in different age groups ($p > 0.05$).
2) Naive-meditators were recruited from the local community and had no previous experience with meditation but expressed interest in learning meditation. Regarding sample homogeneity to the naive meditation practice, sample *t*-test in the NM group demonstrated no significant difference in meditation time in different age groups ($p > 0.05$). Subjects in the control group were familiarized with the meditation instruction for one week before the experiment; they received verbal instructions during the practice (Fucci et al., 2018).

The Research Ethics Committee approved the experiment (n° 1607069/2016), according to the Human Research Ethics Criteria included in the Declaration of Helsinki. The participants signed a Free and Informed Consent Declaration under the ethical standards established in the Helsinki Declaration, 1964.

2.2. Experimental procedure

The participants sat on cushions and rested for 5 min for baseline measurements. The room had acoustic insulation, electrical grounding, and low light.

The participant kept his eyes open during the rest condition. In this context, the EEG acquisition took place in three parts: EEG resting-state 1 (5 min), during the open monitoring meditation (40 min), and EEG resting-state 2 (5 min) (Fig. 1).

An auditory signal marked the beginning and end of each stage. During rest 1 and rest 2, lasting 5 min each, participants sat the usual meditation position in a darkened and noise-free room to minimize sensory interference. The participants were relaxed and seated with their eyes open. Subjects were instructed not to go into a sleep state and relax without entering into a meditative state. Posteriorly, during the meditation, subjects sat in their standard meditation position (i.e., half-lotus position with eyes closed) with a meditation cushion on the floor, and EEG data collection lasted 40 min. After the meditation, the subjects were asked to relax, not meditate, keep their eyes open, and minimize eye movements.

2.3. EEG recording

The electroencephalographic signal was recorded through the 20-channel BrainNet (EMSA Medical Equipment). The silver/silver

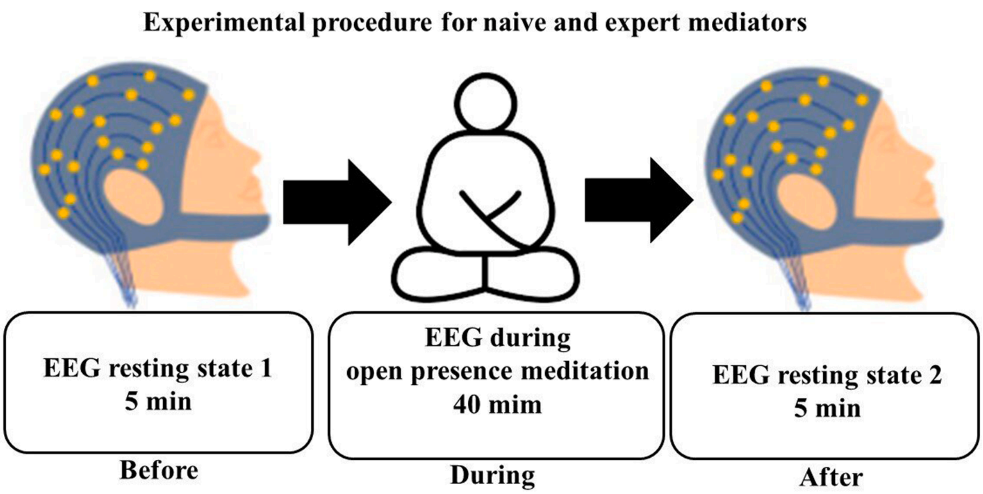

Fig. 1. Illustration of the task protocol for both groups. The experimental procedure was based on EEG resting state before, during and after The Mindful Attention Awareness Scale (MAAS), the trait MAAS evaluates the core characteristic of open monitoring meditation session.

chloride electrodes were positioned through a nylon cap following the international 10–20 system, including binaural reference electrodes. The EEG electrode impedance and electrooculogram (EOG) electrodes were kept below 5kΩ. The acquired data had an amplitude below 100 μV. The sampling rate was 240 Hz. An anti-aliasing low-pass filter with a cut-off frequency of 100 Hz was employed. Its configuration uses 60 Hz Notch digital filtering, with high-pass filters at 0.1 Hz and low pass filters at 40 Hz (Order 2 Butterworth filter), using the Data Acquisition software (Delphi 5.0) developed in the Brain Mapping and Sensorimotor Integration Laboratory.

2.4. EEG data processing

A visual inspection and independent component analysis (ICA) was applied to identify and remove the artifacts in MATLAB version 4.3 (The Mathworks, Inc.) and detect and remove artifacts associated with blinking, vertical, and horizontal EOG and EEG-related fluctuation. We removed the data from individual electrodes that showed contact loss with the scalp or high impedance (>5kΩ). A classical estimator (i.e., parametric, Bartlett Periodogram, using non-overlapping 2 s long [480 samples] rectangular windows) was applied to the Power Spectral Density (PSD), estimated from the Fourier Transform (FT), which was performed using MATLAB.

The EEG coherence is the synchrony in the neural activity of different cortical brain areas picked up by electrophysiological recordings. 10–20 EEG System (Jasper, 1958) is commonly used to study the brain's neural communication and function at rest or during functional tasks. We estimate the coherence (*Coh*) in gamma band (30–100 Hz) by Fast Fourier transform (FFT), using the expression: where G*xy*(*f*) is the cross-spectrum density, and G*xx*(*f*) and G*yy*(*f*) are the spectral densities of the × and y signals (Jasper, 1958). The rectangular windowing started with analysis over long segments on the complete meditation data (whole block analysis) for the derivation pairs Fp1-P3, F3-P3, F7-P3 for the left and Fp2-P4, F4-P4, F8-P4 for the right hemisphere.

The Coh analysis (sometimes called magnitude-squared coherence) between two signals x(*t*) and y(*t*), is a real-valued function that is defined by Fast Fourier transform (FFT), using the formula:

$$Coh^2(f) = \frac{|G_{xy}(f)|^2}{G_{xx}(f)G_{yy}(f)}$$

$|G_{xy}(f)|^2$: Cross-spectrum power.

$G_{xx}(f)G_{yy}(f)$: Spectral power of × and y.

The analysis is adequate in the parameters where Gxy(*f*) is the Cross-spectral density between × and y, and Gxx(*f*) and Gyy(*f*) the auto spectral density of × and y, respectively. |G| is the magnitude of the spectral density. Given the restrictions noted above (ergodicity, linearity), the coherence function estimates the extent to which y(*t*) may be predicted from x(t) by an optimum linear least-squares function. Values of coherence will always satisfy $0 \leq C_{xy}(f) \leq 1$.

2.5. Electrodes of interest

We analyzed the EEG gamma coherence by the EEG gamma absolute power (30 – 100 Hz) in electrode positions on the scalp: Fp1, Fp2, F3, F4, F7, F8, P3, and P4 electrodes. The association between these electrodes demonstrates the sensorimotor integration process among frontal and parietal regions (frontoparietal network), which encodes exteroceptive information and sends them to frontal regions, which will create the motor and cognitive plans. We selected the electrodes based on the functions inbuilt in the perception, attention level, default mode network, executive attention, decision-making, and sensorimotor integration (Adhikari et al., 2013; Hayashi et al., 2014; Balconi and Cobelli, 2015).

2.6. Statistical analysis

The Mauchly's test criteria was used to evaluate the sphericity hypothesis and the Greenhouse-Geisser (G-Gε) procedure to correct degrees of freedom. The normality and homoscedasticity of the data were previously verified by the Levene and Shapiro–Wilk tests.

Two-way ANOVA was applied to identify the EEG gamma coherence differences in each electrode pairing (Fp1-P3, F3-P3, F7-P3, Fp2-P4, F4-P4, F8-P4), with factors: group (EM *vs.* NM) and moment factor (rest 1 *vs.* Meditation *vs.* rest 2) as independent variables. We performed Independent *t*-tests for Group effects and a One-way ANOVA for moment effects to examine the interactions. Correction for multiple comparisons was applied using the Tukey's Test for the six tests conducted (Fp1-P3, F3-P3, F7-P3, Fp2-P4, F4-P4, F8-P4), considering the $p \leq 0.0083$ (0.05/6).

The effect size was estimated as partial eta-squared (η2p). In addition, the analysis effect was evaluated by Cohen's d for Student *t*-test. Statistical power and the 95% confidence interval (95% CI) were calculated for the dependent variables. Statistical power was interpreted as the low power of 0.1 to 0.3; high power from 0.8 to 0.9. The effect of magnitude was interpreted using the recommendations suggested by Cohen (1998): insignificant < 0.19; small from 0.20 to 0.49; mean from 0.50 to 0.79; large from 0.80 to 1.29. The probability of 5% for type I error was adopted for all analyzes ($p < 0.05$). The analyses were conducted in SPSS for Windows version 20.0 (SPSS Inc., Chicago, Il, USA).

3. Results

Two-way ANOVAs did not show a significant result for F3-P3 and F7-P3 electrode pairs ($p > 0.0083$). However, a two-way ANOVA showed interaction between group and moment for Fp1-P3 electrode pair: $F(2, 1962) = 29.12$; $p = 0.001$; $\eta^2p = 0.55$; power = 100% (Fig. 2). Analyzing the interaction within each group by One-way ANOVA showed significant results only for EM group [$F(2,1027) = 45.76$; $p = 0.002$; $\eta^2p = 0.55$; power = 100%]. We observed that the frontoparietal EEG gamma coherence decreased during meditation in relation EEG resting state. The analyze between groups by Independent *t*-tests showed higher coherence to EM in the rest 1 in relation NM $t(159) = 5.73$; $p = 0.001$; $d = 0.52$; CI95% 1.22 – 1.48, and rest 2 $t(161) = 4.00$; $p = 0.0002$; $d = 0.56$; CI95% 1.52 – 1.79.

The two-way ANOVA for Fp2-P4 electrode pair showed interaction between group and moment, with $F(2, 1992) = 6.26$; $p = 0.001$; $\eta^2p = 0.62$; power = 98% (Fig. 3). When analyzing the interaction by one-way ANOVA within each group, significant results were detected only for EM [$F(2,1043) = 16.59$; $p = 0.001$; $\eta^2p = 0.57$; power = 100%]. The rest 1 was higher than meditation and rest 2. The analysis between groups by Independent *t*-tests showed higher coherence to EM in the rest 1 in relation to NM $t(164) = 2,87$; $p = 0.001$; $d = 0.55$; CI95% 1.78 – 1.92.

The two-way ANOVA demonstrated interaction between group and moment for F4-P4 electrode pair: $F(2, 1985) = 17.01$; $p = 0.0001$; $\eta^2p = 0.52$; power = 100% (Fig. 4). Analyzing the interaction within each group by One-way ANOVA showed significant result only for EM $F(2,1040) = 6.92$; $p = 0.0001$; $\eta^2p = 0.51$; power = 98%. The meditation was higher than rest 1 and rest 2. The differences between groups by Independent *t*-tests showed higher coherence for rest 1 in EM $t(162) = 4.74$; $p = 0.002$; $d = 0.50$; CI95% 1.56 – 1.70, and differences in the meditation state $t(1659) = 5.95$; $p = 0.0001$; $d = 0.51$; CI95% 1.54 – 1.81.

A two-way ANOVA showed interaction between group and moment for F8-P4 electrode pair: $F(2, 1986) = 29.12$; $p = 0.0002$; $\eta^2p = 0.53$; power = 99% (Fig. 5). When analyzing the interaction within each group by One-way ANOVA, we showed significant result only for EM $F(2,1042) = 18.62$; $p = 0.001$; $\eta^2p = 0.55$; power = 100%. The frontoparietal EEG gamma coherence increased during meditation. Independent *t*-tests between groups demonstrated differences for rest 1 $t(164) = 5.88$; $p = 0.0018$; $d = 0.43$; CI95% 1.72 – 1.85, meditation $t(1661) = 5.79$; $p = 0.001$; $d = 0.48$; CI95% 1.98 – 2.03, and rest 2 $t(161) = 3.92$; $p = 0.002$; $d = 0.52$; CI95% 2.36 – 2.98, which demonstrates an increase in coherence in the EM group.

In summary, the main results are comparisons between groups, especially with differences during the meditative state.

4. Discussion

We analyzed the differences in the EEG gamma coherence in the frontoparietal network before, during, and after open monitoring meditation in experts and naïve meditators. However, the findings showed a small effect during the comparisons. The hypothesis was that regular open monitoring meditation modulates the frontoparietal network by EEG gamma-band coupling related to the time practice effect. We included two resting states, one before and another after meditation, in the experimental design to control the experimental variables more. We did not expect to find statistical differences between them.

Firstly, compared to NM group, EM group showed higher frontoparietal EEG gamma coherence in all resting-state 1, which may be related to the training of EM group for meditation. In addition, the meditation moment has a mild to moderate increase for F4-P4 and F8-P4 electrode pairs. In this context, the activity in the right frontoparietal network is related to attention in visual detection (Quentin et al., 2015; Olesen, Westerberg, and Klingberg, 2004), self-regulation of attention and emotion (Vago and Silbersweig, 2012), and sensorimotor integration during meditation (Havranek et al., 2012; Blanke, Slater, and Serino, 2015). The EEG gamma coherence

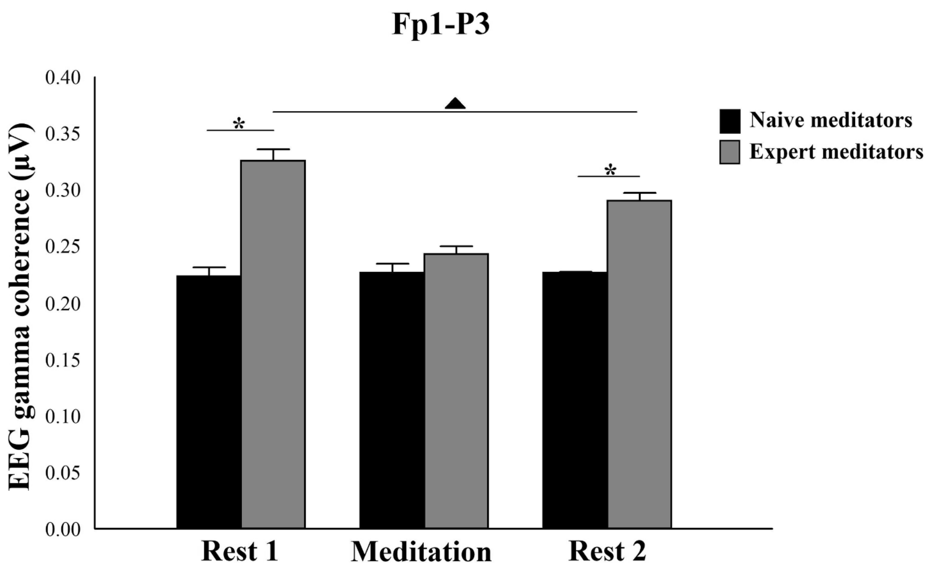

Fig. 2. Differences for EEG gamma coherence in relation to Fp1-P3 electrode pairs. The EEG resting state before, during and after the meditation showed differences in the gamma coherence for experts meditator. Statistical differences in the within group are represented by the asterisk (▲). In addition, comparisons between groups showed differences for rest 1 and rest 2. The graphic is represent by the mean ± standard error and the statistically significant differences ($p \leq 0.0083$) are indicated with (*).

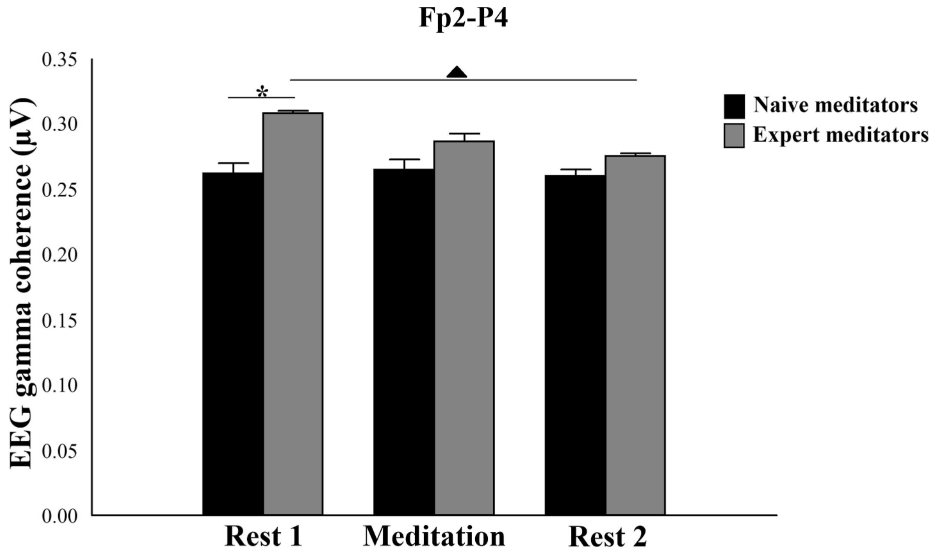

Fig. 3. Differences for EEG gamma coherence in relation to Fp2-P4 electrode pairs. The analysis within group for experts meditator showed differences in the EEG gamma coherence between rest 1 and meditation, as well as rest 1 and rest 2. Statistical differences in the within group are represented by the asterisk (▲). In addition, comparisons between groups showed differences for rest 1. The graphic is represent by the mean ± standard error and the statistically significant differences ($p \leq 0.0083$) are indicated with (*).

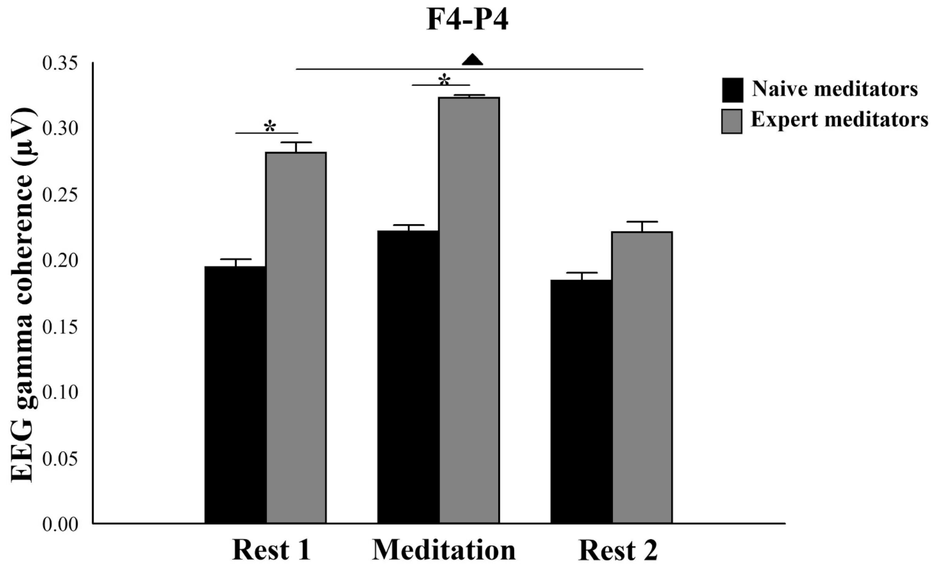

Fig. 4. Differences for EEG gamma coherence in relation to F4-P4 electrode pairs. The EEG resting state before, during and after the meditation showed differences in the gamma coherence for experts meditator. Statistical differences in the within group are represented by the asterisk (▲). In addition, comparisons between groups showed differences for rest 1 and meditation. The graphic is represent by the mean ± standard error and the statistically significant differences ($p \leq 0.0083$) are indicated with (*).

increase over electrodes associated with the frontoparietal network can indicate an attentional skill for arising interoceptive and external stimuli. Other studies found similar results (Benchenane, Tiesinga, and Battaglia, 2011; Berkovich-Ohana, Glicksohn, and Goldstein, 2012). The gamma-band is associated with improving cognitive abilities (Miltner et al., 1999; Babiloni et al., 2006; Travis and Shear, 2010; Cavinato et al., 2015) and body and perceptual self-consciousness (Naito and Morita, 2014). Moreover, the dysfunction of this network is associated with mind/body representation disorder (Giummarra et al., 2011), such as schizophrenia (Velasques et al., 2013) and disorder of consciousness (Cavinato et al., 2015).

We also observed a frontoparietal EEG gamma coherence decrease during meditation concerning EEG resting state for Fp2-P4 and Fp1-P3. Previous studies demonstrated a higher frontoparietal gamma in OM meditative processes (Lutz et al., 2004; Lee et al., 2018). Unlike most studies, where there is a comparison of frontoparietal gamma between experienced meditators and controls, we found a

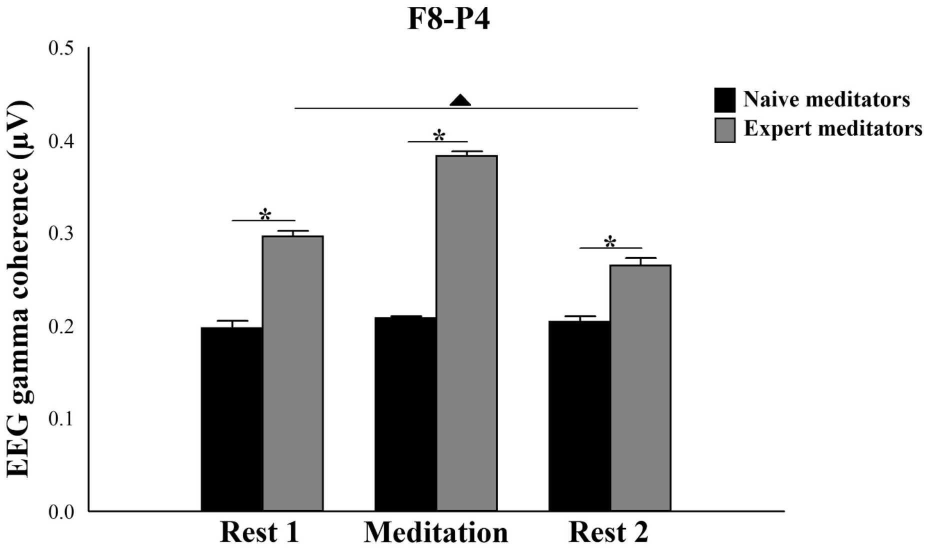

Fig. 5. Differences for EEG gamma coherence in relation to F8-P4 electrode pairs. The EEG resting state before, during and after the meditation showed differences in the gamma coherence for experts meditator. Statistical differences in the within group are represented by the asterisk (▲). In addition, comparisons between groups showed differences for rest 1, meditation and rest 2. The graphic is represent by the mean ± standard error and the statistically significant differences ($p \leq 0.0083$) are indicated with (*).

difference within experienced meditators themselves: during meditation, they show a decrease in frontoparietal gamma when compared to rest. While the functional role of gamma activity is not yet clear, one hypothesis is that it induces neuroplasticity via repetition, with practice, the process to focus attention becomes less effortful (Brefczynski-Lewis et al., 2007; Babiloni et al., 2010; Tang, Hölzel, and Posner, 2015; Russell, 2017). Then, during meditation, the EM group demands less participation of the frontoparietal network, reaching a state of relaxation and demanding less communication between these areas.

On the other hand, we can suggest that increase in right EEG gamma coherence can indicate the time meditation practice is a relevant factor to define which pattern of attention of the practitioners (narrow or wide) (Jha, Krompinger, and Baime, 2007). This pattern has a specific activity in the brain areas in a particular way, mainly in increasing awareness through the intent not to analyze, not judge, and not expect anything (Tei et al., 2009). We observed that the meditation training promotes control state sensorium differently, modulating the EEG gamma coherence amplitude in expert mediators groups. Thus, EM group showed an EEG pattern promoted by a long-term training, tending to effortless attention caused by moderate increase coupling between frontal and parietal areas during mindfulness compared to NM. This pattern improves the experience of multi-sensory processing (Vago, 2014; Blanke, Slater, and Serino, 2015). On the other hand, the NM showed no difference between all moments and presented lower frontoparietal EEG gamma coherence. These results support the hypothesis that meditation training is associated with developing the neural networks necessary for improving attention in a particular way (Cahn, Delorme, and Polich, 2010; Tang et al., 2019).

The findings in the right frontoparietal area can represent the relationship between attention and practice level (Slagter, Davidson, and Lutz, 2011; Saggar et al., 2012), with lower frontal cortical demonstrating broad attention (Chiesa, Serretti, and Jakobsen, 2013; Tanaka et al., 2014). Our results are similar, where the EM had shown greater frontoparietal EEG gamma coherence when compared to NM. Thus, we suggest that the EEG gamma coherence is a neurophysiological biomarker of the frontoparietal coupling during open monitoring meditation (Bishop et al., 2004; Hauswald et al., 2015), in focus, the gamma coherence in the right hemisphere may be engaged in top-down controlled sensory awareness, whereas the left hemisphere may be responsible for directing motor attention in relation to voluntary initiation and control of behavior (Yordanova et al., 2021). The results further suggest that attentional effort control can be guided by either the right hemisphere depending on the level of attentional training and/or simultaneously activated processes. Interestingly, from a neuro-functional perspective, this result may reflect facilitation of conscious access, and a transference of attention-dependent conscious access to higher-frequency networks in the brain (Vago, 2014). However, we should be cautious with this interpretation because of the limitation of the NM group's little experience with open monitoring meditation (Chiesa, Serretti, and Jakobsen, 2013).

A major finding is higher right frontoparietal coupling in EM group when compared to NM group. However, since it is a cross-sectional study, intervention effect should be addressed. We suggest that the increased coupling in right hemisphere during EEG resting state and meditation is a trait of EM regular training. This trait underlies the attentional process, with reduced interference (i.e. inner speech, self-narrative, thoughts) and non-judgment perception (Farb et al., 2007; Brown and Jones, 2010; Cahn, Delorme, and Polich, 2013). Furthermore, the enhanced coupling in the right frontoparietal is consistent with increased mutual information that may underlie the cognitive functions of expert mediators (Gupta and Bahmer, 2019).

This study presents some limitations: the sample size; however, the effects test and statistical power in the analysis decrease the possibility of a type II error. Another limitation is that the EM group has a deeper impact on predictive processes and improves gamma coherence of the frontoparietal network during EEG resting state, compared to a NM group. An important limitation was not to explore the role of executive brain processes in sustaining different forms of meditation as a function of Inter-hemisphere coherence. These results could demonstrate compared to unexperienced practitioners, meditation states in long-term practitioners induced highly specific connectivity patterns of fronto-parietal networks. This observation generally can indicate that the executive processes of attentional control and cognitive monitoring have a specific role in supporting brain states of meditation.

An important limitation was the lack of an assessment of the receptive state of mind by The Mindful Attention Awareness Scale (MAAS); the trait MAAS evaluates the core characteristic of mindfulness, namely, a receptive state of mind in which attention, informed by a sensitive awareness of what is occurring in the present. Additionally, subjects' compliance with instruction to avoid movements was not confirmed independently by electromyography (EMG).

Significant limitations related to the EEG recording are present in our study, which reduce the inferential power, such as no inter-hemispheric coherence analysis, which could present the more specific functionality of the attention network. Due to the absence of P7 and P8 electrodes, the current findings may not accurately reflect the dorsal attention network. However, the study contributes to the state of the art and future research that explores the mechanisms underlying the modulation of the cortical coupling and neural efficiency by meditation by implementing trial-by-trial analysis of the auditory responses and other stimuli.

5. Conclusion

Our findings underscore the importance of frontoparietal EEG gamma coherence in understanding the neurophysiology of awareness, concerning open meditation. The long-lasting training improves the coupling in the right frontoparietal network during sensorial synchrony and stimuli perception. The increase in EEG gamma coherence in the EM group highlights a putative link between gamma oscillatory activity in meditation and attentional processes (e.g., reorienting attention from auditory and visual stimuli to the main task-set of the practice), which will foster the understanding of the brain mechanisms involved in the mechanisms of meditation practices in clinical settings.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.concog.2022.103354.

References

Adhikari, B. M., Goshorn, E. S., Lamichhane, B., & Dhamala, M. (2013). Temporal-order judgment of audiovisual events involves network activity between parietal and prefrontal cortices. *Brain Connect., 3*(5), 536–545.

Babiloni, C., Marzano, N., Infarinato, F., Iacoboni, M., Rizza, G., Aschieri, P., et al. (2010). "Neural efficiency" of experts' brain during judgment of actions: A high-resolution EEG study in elite and amateur karate athletes. *Behav Brain Res., 207*(2), 466–475.

Babiloni, C., Vecchio, F., Cappa, S., Pasqualetti, P., Rossi, S., Miniussi, C., et al. (2006). Functional frontoparietal connectivity during encoding and retrieval processes follows HERA model. *A high-resolution study. Brain Res Bull., 68*(4), 203–212.

Balconi, M., & Cobelli, C. (2015). rTMS on left prefrontal cortex contributes to memories for positive emotional cues: A comparison between pictures and words. *Neuroscience., 26*(287), 93–103.

Benchenane, K., Tiesinga, P. H., & Battaglia, F. P. (2011). Oscillations in the prefrontal cortex: A gateway to memory and attention. *Curr Opin Neurobiol., 21*(3), 475–485.

Bendat, J. S., & Allan, G. P. (2011). *Random data: Analysis and measurement procedures* (p. 729). John Wiley & Sons.

Berkovich-Ohana, A., Glicksohn, J., & Goldstein, A. (2012). Mindfulness-induced changes in gamma band activity - implications for the default mode network, self-reference and attention. *Clin Neurophysiol., 123*(4), 700–710.

Bishop, S., Duncan, J., Brett, M., & Lawrence, A. D. (2004). Prefrontal cortical function and anxiety: Controlling attention to threat-related stimuli. *Nat Neurosci., 7*(2), 184–188.

Blanke, O., Slater, M., & Serino, A. (2015). Behavioral, Neural, and Computational Principles of Bodily Self-Consciousness. *Neuron., 88*(1), 145–166.

Brefczynski-Lewis, J. A., Lutz, A., Schaefer, H. S., Levinson, D. B., & Davidson, R. J. (2007). Neural correlates of attentional expertise in long-term meditation practitioners. Version 2. *Proc Natl Acad Sci U S A., 104*(27), 11483–11488.

Brown, C. A., & Jones, A. K. (2010). Meditation experience predicts less negative appraisal of pain: Electrophysiological evidence for the involvement of anticipatory neural responses. *Pain., 150*(3), 428–438.

Cahn, B. R., Delorme, A., & Polich, J. (2010). Occipital gamma activation during Vipassana meditation. *Cogn Process., 11*(1), 39–56.

Cavinato, M., Genna, C., Manganotti, P., Formaggio, E., Storti, S. F., Campostrini, S., et al. (2015). Coherence and Consciousness: Study of Fronto-Parietal Gamma Synchrony in Patients with Disorders of Consciousness. *Brain Topogr., 28*(4), 570–579.

Chambers, R., Gullone, E., & Allen, N. B. (2009). Mindful emotion regulation: An integrative review. *Clin Psychol Rev., 29*(6), 560–572.

Chica, A. B., & Bartolomeo, P. (2012). Attentional routes to conscious perception. *Front Psychol., 18*(3), 1.

Chiesa, A., Serretti, A., & Jakobsen, J. C. (2013). Mindfulness: Top-down or bottom-up emotion regulation strategy? *Clin Psychol Rev., 33*(1), 82–96.

Chiesa, A., & Serretti, A. (2009). Mindfulness-based stress reduction for stress management in healthy people: A review and meta-analysis. *J Altern Complement Med., 15*(5), 593–600.

Cohen, J. (1988). *Statistical Power Analysis for the Behavioral Sciences*. New York, NY: Routledge Academic.

Davis, D. M., & Hayes, J. A. (2011). What are the benefits of mindfulness? A practice review of psychotherapy-related research. *Psychotherapy, 48*(2), 198–208. https://doi.org/10.1037/a0022062

Dentico, D., Ferrarelli, F., Riedner, B. A., Smith, R., Zennig, C., Lutz, A., et al. (2016). Short Meditation Trainings Enhance Non-REM Sleep Low-Frequency Oscillations. *PLoS One., 11*(2), Article e0148961.
Dor-Ziderman, Y., Berkovich-Ohana, A., Glicksohn, J., & Goldstein, A. (2013). Mindfulness-induced selflessness: A MEG neurophenomenological study. *Front Hum Neurosci., 7*, 582.
Eberth, J., & Sedlmeier, P. (2012). The effects of mindfulness meditation: A meta-analysis. *Mindfulness., 3*, 174–189.
Farb, N. A., Segal, Z. V., Mayberg, H., Bean, J., McKeon, D., Fatima, Z., et al. (2007). Attending to the present: Mindfulness meditation reveals distinct neural modes of self-reference. *Soc Cogn Affect Neurosci., 2*(4), 313–322.
Ferrarelli, F., Smith, R., Dentico, D., Riedner, B. A., Zennig, C., Benca, R. M., et al. (2013). Experienced mindfulness meditators exhibit higher parietal-occipital EEG gamma activity during NREM sleep. *PLoS One.* 28;8(8):e73417.
Florian Kurth, Allan MacKenzie-Graham, Arthur W. Toga, Eileen Luders (2015). Shifting brain asymmetry: the link between meditation and structural lateralization, Social Cognitive and Affective Neuroscience, Volume 10, Issue 1, January 2015, Pages 55–61, https://doi.org/10.1093/scan/nsu029.
Fucci, E., Abdoun, O., Caclin, A., Francis, A., Dunne, J. D., Ricard, M., et al. (2018). Differential effects of non-dual and focused attention meditations on the formation of automatic perceptual habits in expert practitioners. *Neuropsychologia., 119*, 92–100.
Giummarra, M. J., Bradshaw, J. L., Nicholls, M. E., Hilti, L. M., & Brugger, P. (2011). Body integrity identity disorder: Deranged body processing, right fronto-parietal dysfunction, and phenomenological experience of body incongruity. *Neuropsychol Rev., 21*(4), 320–333.
Gupta, D. S., & Bahmer, A. (2019). Increase in mutual information during interaction with the environment contributes to perception. *Entropy, 21*, 365.
Hasenkamp, W., Wilson-Mendenhall, C. D., Duncan, E., & Barsalou, L. W. (2012). Mind-wandering and attention during focused meditation: A fine-grained temporal analysis of fluctuating cognitive states. *Neuroimage., 59*(1), 750–760.
Hauswald, A., Übelacker, T., Leske, S., & Weisz, N. (2015). What it means to be Zen: Marked modulations of local and interareal synchronization during open-monitoring meditation. *Neuroimage., 108*, 265–273.
Havranek, M., Langer, N., Cheetham, M., & Jäncke, L. (2012). Perspective and agency during video gaming influences spatial presence experience and brain activation patterns. *Behav Brain Funct., 8*, 34.
Hayashi, M. J., Kantele, M., Walsh, V., Carlson, S., & Kanai, R. (2014). Dissociable neuroanatomical correlates of subsecond and supra second time perception. *J Cogn Neurosci., 26*(8), 1685–1693.
Hofmann, S. G., Sawyer, A. T., Witt, A. A., & Oh, D. (2010). The effects of mindfulness-based therapy on anxiety and depression: A meta-analytic review. *J. Consult. Clin. Psychol., 78*, 169–183.
Hölzel, B. K., Carmody, J., Vangel, M., Congleton, C., Yerramsetti, S. M., Gard, T., et al. (2011). Mindfulness practice leads to increases in regional brain gray matter density. *Psychiatry Res., 191*(1), 36–43.
Hudak, J., Hanley, A. W., Marchand, W. R., Nakamura, Y., Yabko, B., & Garland, E. L. (2021). Endogenous theta stimulation during meditation predicts reduced opioid dosing following treatment with Mindfulness-Oriented Recovery Enhancement. Neuropsychopharmacology: Official publication of the American College of. *Neuropsychopharmacology, 46*(4), 836–843.
Jha, A. P., Krompinger, J., & Baime, M. J. (2007). Mindfulness training modifies subsystems of attention. *Cogn Affect Behav Neurosci., 7*(2), 109–119.
Jo, H. G., Hinterberger, T., Wittmann, M., & Schmidt, S. (2015). Do meditators have higher awareness of their intentions to act? *Cortex., 65*, 149–158.
Kerr, C. E., Sacchet, M. D., Lazar, S. W., Moore, C. I., & Jones, S. R. (2013). Mindfulness starts with the body: Somatosensory attention and top-down modulation of cortical alpha rhythms in mindfulness meditation. *Front Hum Neurosci., 13*(7), 12.
Khoury, B., Lecomte, T., Fortin, G., Masse, M., Therien, P., Bouchard, V., et al. (2013). Mindfulness-Based Therapy: A Comprehensive Meta-Analysis. *Clin. Psychol. Rev., 33*, 763–771.
Kumari, V., Antonova, E., Wright, B., Hamid, A., Hernandez, E. M., Schmechtig, A., et al. (2017). The mindful eye: Smooth pursuit and saccadic eye movements in meditators and non-meditators. *Conscious Cogn., 48*, 66–75.
Kummar, A. S., Correia, H., & Fujiyama, H. (2019). A Brief Review of the EEG Literature on Mindfulness and Fear Extinction and its Potential Implications for Posttraumatic Stress Symptoms (PTSS). *Brain Sci.* 27;9(10):258.
Langer, N., von Bastian, C. C., Wirz, H., Oberauer, K., & Jäncke, L. (2013). The effects of working memory training on functional brain network efficiency. *Cortex., 49*(9), 2424–2438.
Lee, D. J., Kulubya, E., Goldin, P., Goodarzi, A., & Girgis, F. (2018). Review of the Neural Oscillations Underlying Meditation. *Front. Neurosci., 12*, 178. https://doi.org/10.3389/fnins.2018.00178
Lomas, T., Ivtzan, I., & Fu, C. H. (2015). A systematic review of the neurophysiology of mindfulness on EEG oscillations. *Neurosci Biobehav Rev., 57*, 401–410.
Lutz, A., Greischar, L. L., Rawlings, N. B., Ricard, M., & Davidson, R. J. (2004). Long-term meditators self-induce high-amplitude gamma synchrony during mental practice. *Proc. Natl. Acad. Sci. U.S.A., 101*, 16369–16373. https://doi.org/10.1073/pnas.0407401101
Lutz, A., Slagter, H. A., Dunne, J. D., & Davidson, R. J. (2008). Attention regulation and monitoring in meditation. *Trends Cogn Sci., 12*(4), 163–169.
Malinowski, P. (2013). Neural mechanisms of attentional control in mindfulness meditation. *Front Neurosci., 4*(7), 8.
Miltner, W. H., Braun, C., Arnold, M., Witte, H., & Taub, E. (1999). Coherence of gamma-band EEG activity as a basis for associative learning. *Nature., 397*(6718), 434–436.
Naito, E., & Morita, T. (2014). Neural representation of human body schema and corporeal self-consciousness. *Brain Nerve., 66*(4), 367–380.
Oldfield, R. C. (1971). The assessment and analysis of handedness: The Edinburgh inventory. *Neuropsychologia., 9*(1), 97–113.
Olesen, P. J., Westerberg, H., & Klingberg, T. (2004). Increased prefrontal and parietal activity after training of working memory. *Nat Neurosci., 7*(1), 75–79.
Quentin, R., Chanes, L., Vernet, M., & Valero-Cabré, A. (2015). Fronto-Parietal Anatomical Connections Influence the Modulation of Conscious Visual Perception by High-Beta Frontal Oscillatory Activity. *Cereb Cortex., 25*(8), 2095–2101.
Russell, T. (2017). *What is Mindfulness?* (1st ed., p. 144 p.). Watkins Publishing.
Saggar, M., King, B. G., Zanesco, A. P., Maclean, K. A., Aichele, S. R., Jacobs, T. L., et al. (2012). Intensive training induces longitudinal changes in meditation state-related EEG oscillatory activity. *Front Hum Neurosci., 10*(6), 256.
Scheibner, H. J., Bogler, C., Gleich, T., Haynes, J. D., & Bermpohl, F. (2017). Internal and external attention and the default mode network. *Neuroimage., 148*, 381–389.
Slagter, H. A., Davidson, R. J., & Lutz, A. (2011). Mental training as a tool in the neuroscientific study of brain and cognitive plasticity. *Front Hum Neurosci., 10*(5), 17.
Tanaka, G. K., Maslahati, T., Gongora, M., Bittencourt, J., Lopez, L. C., Demarzo, M. M., et al. (2015). Effortless Attention as a Biomarker for Experienced Mindfulness Practitioners. *PLoS One., 10*(10), Article e0138561.
Tanaka, G. K., Peressutti, C., Teixeira, S., Cagy, M., Piedade, R., Nardi, A. E., et al. (2014). Lower trait frontal theta activity in mindfulness meditators. *Arq Neuropsiquiatr., 72*(9), 687–693.
Tang, Y. Y., Hölzel, B. K., & Posner, M. I. (2015). The neuroscience of mindfulness meditation. *Nat Rev Neurosci., 16*(4), 213–225.
Tang, Y. Y., Tang, R., Rothbart, M. K., & Posner, M. I. (2019). Frontal theta activity and white matter plasticity following mindfulness meditation. *Curr Opin Psychol., 28*, 294–297.
Tei, S., Faber, P. L., Lehmann, D., Tsujiuchi, T., Kumano, H., Pascual-Marqui, R. D., et al. (2009). Meditators and non-meditators: EEG source imaging during resting. *Brain Topogr., 22*(3), 158–165.
Travis, F., & Shear, J. (2010). Focused attention, open monitoring and automatic self-transcending: Categories to organize meditations from Vedic. *Buddhist and Chinese traditions. Conscious Cogn., 19*(4), 1110–1118.

Vago, D. R. (2014). Mapping modalities of self-awareness in mindfulness practice: A potential mechanism for clarifying habits of mind. *Ann N Y Acad Sci., 1307*, 28–42.

Velasques, B., Bittencourt, J., Diniz, C., Teixeira, S., Basile, L. F., Inácio Salles, J., et al. (2013). Changes in saccadic eye movement (SEM) and quantitative EEG parameter in bipolar patients. *J Affect Disord., 145*(3), 378–385.

Yordanova, J., Kolev, V., Nicolardi, V., et al. (2021). Attentional and cognitive monitoring brain networks in long-term meditators depend on meditation states and expertise. *Sci Rep, 11*, 4909. https://doi.org/10.1038/s41598-021-84325-3